

MIDTERM EXAM-ORAL MAKEUP EXAM

ECO403/503–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

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Problem 1: Problem 1: Course overview, summary, and interrelationships

Problem 2: Hat puzzle

Problem 3: NU Roommates

You (player 1) have a roommate (player 2). Your NU dormitory room is dirty. Cleaning it requires an effort cost of $c > 0$. Will either of you clean it? Each of you has two available actions: Clean (C) or go to the Movies (M).

Each player has two possible types, representing their valuation for a clean room: v_H (high value) and $v_L = 0$ (low value). The probability of being the high type is $p \in (0, 1)$.

Assume the following conditions on the parameters:

$$v_H > c \quad \text{and} \quad v_H(1 - p) < c.$$

The payoffs of a player with valuation v are given by:

$$u_i(a_i, a_j, v) = \begin{cases} v - c & \text{if } a_i = C, a_j \in \{C, M\}, \\ v & \text{if } a_i = M, a_j = C, \\ 0 & \text{if } a_i = a_j = M. \end{cases}$$

Remark: The payoff is the same whether both of you clean or only one of you does. If either player cleans, they bear the full cost c .

Suppose now that the payoffs for a player with valuation v are:

$$u_i(a_i, a_j, v) = \begin{cases} v - c/2 & \text{if } a_i = a_j = C, \\ v - c & \text{if } a_i = C, a_j = M, \\ v & \text{if } a_i = M, a_j = C, \\ 0 & \text{if } a_i = a_j = M. \end{cases}$$

Problem 4: Sharing information

Suppose that there are two players, 1, and 2. The set of states is $\Omega = \{1, 2, \dots, 12\}$ and the common prior of the two players is $\mu(\omega)$. Assume $\mu(\omega) > 0$ for all $\omega \in \Omega$.

Each player i observes, privately, the realization of the random variable $X_i : \Omega \rightarrow \mathbb{R}$, where these random variables are defined as follows:

$$X_1(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2, 3, 4\} \\ 2 & \text{if } \omega \in \{5, 6, 7, 8\} \\ 3 & \text{if } \omega \in \{9, 10, 11, 12\} \end{cases} \quad X_2(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2\} \\ 2 & \text{if } \omega \in \{3, 4\} \\ 3 & \text{if } \omega \in \{5, 6\} \\ 4 & \text{if } \omega \in \{7, 8\} \\ 5 & \text{if } \omega \in \{9, 10\} \\ 6 & \text{if } \omega \in \{11, 12\} \end{cases}$$

Let $A \in 2^\Omega$ be an event. Define $Q_i^0(\omega) = E[\mathbb{1}_A | X_i](\omega)$ for every $\omega \in \Omega$, where $\mathbb{1}_A$ is the indicator function for event A . Then, for $t = 1, 2, \dots$, define, for every $\omega \in \Omega$:

$$Q_1^t(\omega) = E[\mathbb{1}_A | X_1, Q_2^0, \dots, Q_2^{t-1}](\omega), \quad Q_2^t(\omega) = E[\mathbb{1}_A | X_2, Q_1^0, \dots, Q_1^{t-1}](\omega).$$

Answer the following True/False questions. For the answer to be valid, you should provide an argument/proof. Hint: All proofs here are very short.

- 4.1. True or False. For any μ and any event A , $\lim_{t \rightarrow \infty} Q_i^t$ exists for $i = 1, 2$ and the limits are equal.
- 4.2. True or False. For any μ and any event A , the players always end up with the same information partitions at the end of the process.

Problem 5: A proverb about knowledge

Consider the following Arabian proverb:

There are four sorts of men:

He who knows not and knows not he knows not: he is a fool—shun him;

He who knows not and knows he knows not: he is simple—teach him;

He who knows and knows not he knows: he is asleep—wake him;

He who knows and knows he knows: he is wise—follow him.

Define a *possibility correspondence* as a function $P : \Omega \rightarrow 2^\Omega$ that is not necessarily partitional. Define its induced knowledge operator as usual

$$KA = \{\omega \in \Omega : P(\omega) \subseteq A\}.$$

Consider $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ and the following possibility correspondence:

$$P(\omega_1) = \{\omega_1, \omega_2\}$$

$$P(\omega_2) = \{\omega_2, \omega_3\}$$

$$P(\omega_3) = \{\omega_3\}$$

$$P(\omega_4) = \{\omega_4\}$$

Consider an agent whose knowledge operator K is induced by the previous possibility correspondence. Can this agent be the fool? Can this agent be the simple? Can this agent be asleep? Can this agent be wise?